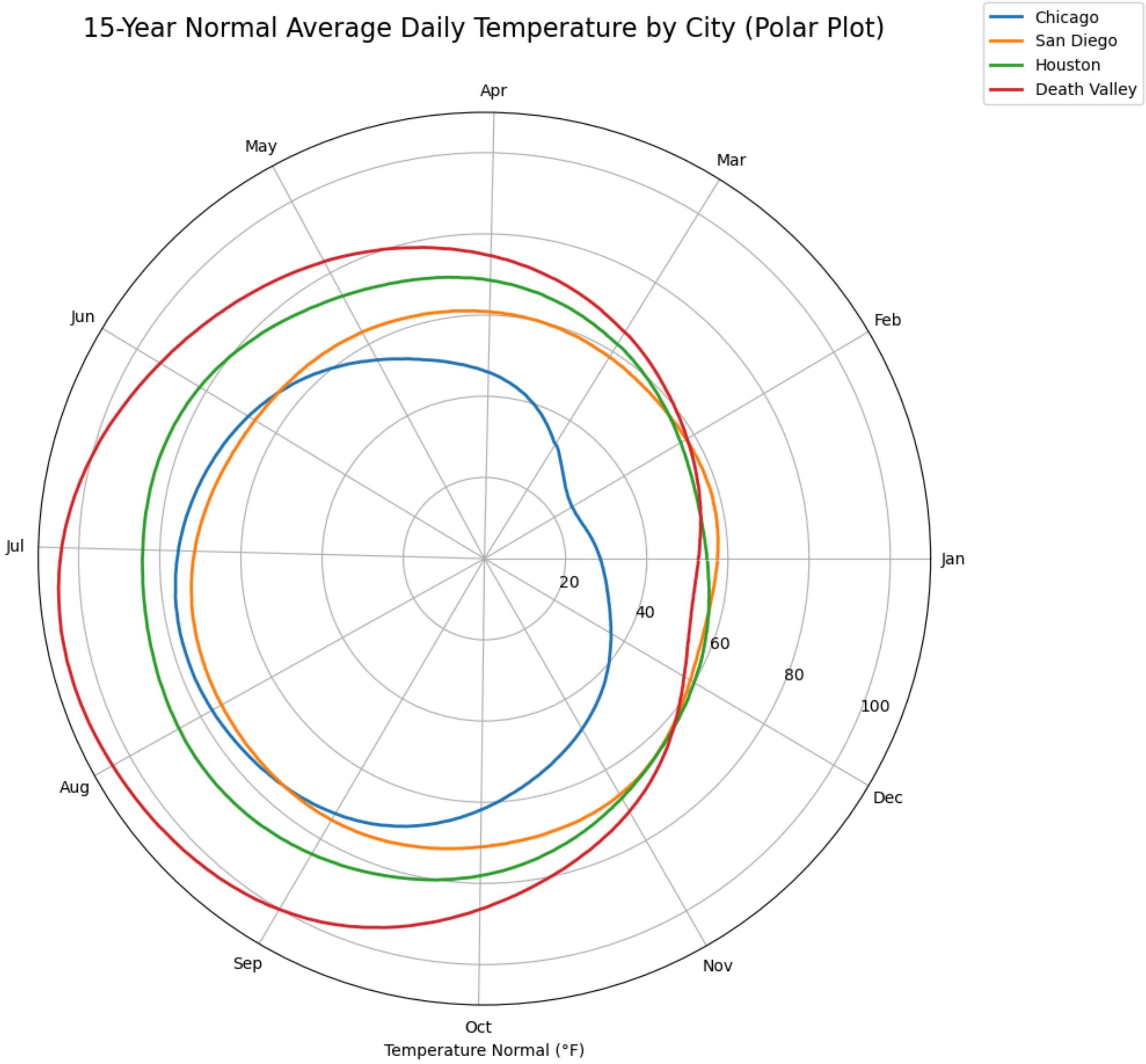
Question 1

In [8]:
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

file_data = {
    "Chicago": "USW00014819.csv",
    "San Diego": "USW00093107.csv",
    "Houston": "USW00012918.csv",
    "Death Valley": "USC00042319.csv"
}
TEMP_COLUMN = "DLY-TAVG-NORMAL"
MAX_DAYS = 365

plt.figure(figsize=(10, 10))
ax = plt.subplot(111, polar=True)

for city, file_name in file_data.items():
    try:
        df = pd.read_csv(file_name)
        temps = df[TEMP_COLUMN].astype(float)

        theta = np.linspace(0, 2 * np.pi, len(temps), endpoint=False)

        ax.plot(theta, temps, label=city, linewidth=2)

    except FileNotFoundError:
        print(f"File not found for {city}: {file_name}")
        continue
    except KeyError:
        print(f"Column '{TEMP_COLUMN}' not found in file for {city}.")
        continue

ax.set_title("15-Year Normal Average Daily Temperature by City (Polar Plot)'

month_days = [0, 31, 59, 90, 120, 151, 181, 212, 243, 273, 304, 334]
month_names = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep
angles = [d * 2 * np.pi / MAX_DAYS for d in month_days]

ax.set_xticks(angles)
ax.set_xticklabels(month_names, fontsize=10)

ax.set_rlabel_position(-22.5)
ax.set_xlabel("Temperature Normal (°F)")

ax.legend(loc='lower left', bbox_to_anchor=(1.05, 1))

plt.show()
```



The above chart shows the average temperature between 4 cities. This chart makes it easier to compare temperature changes with the seasons.
Chicago (blue color) shows lowest temperature during Nov to Apr among all cities and Death Valley (red color) shows highest temperature during May to Sept.

Question 2

https://app.powerbi.com/groups/me/reports/6a2e0803-7c30-417f-bf76-1d9a30040f17/ece6ddc3a289f652bdd3?experience=power-bi

In [ ]:

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Question 3

Tableau vs Power BI

```
Functionality:
Power BI uses Power Query (M) for data shaping and supports
row-level security (RLS) in the model/service.
Tableau provides rich interaction design via Parameter
```

```
Actions and Viz in Tooltip for layered drill-ins without
leaving the view.
In Tableau, visuals customization and smooth animation stand
out, while Power BI is stronger for integration with Excel,
data cleaning and fast filtering between visuals.
```

https://app.powerbi.com/groups/e42477f9-1a76-47a5-a73a-9368312862cd/reports/ee702150-f34b-40d4-850f-f4eb4ee03ba4/aad20779cdb901e3f194?experience=power-bi

In [ ]: